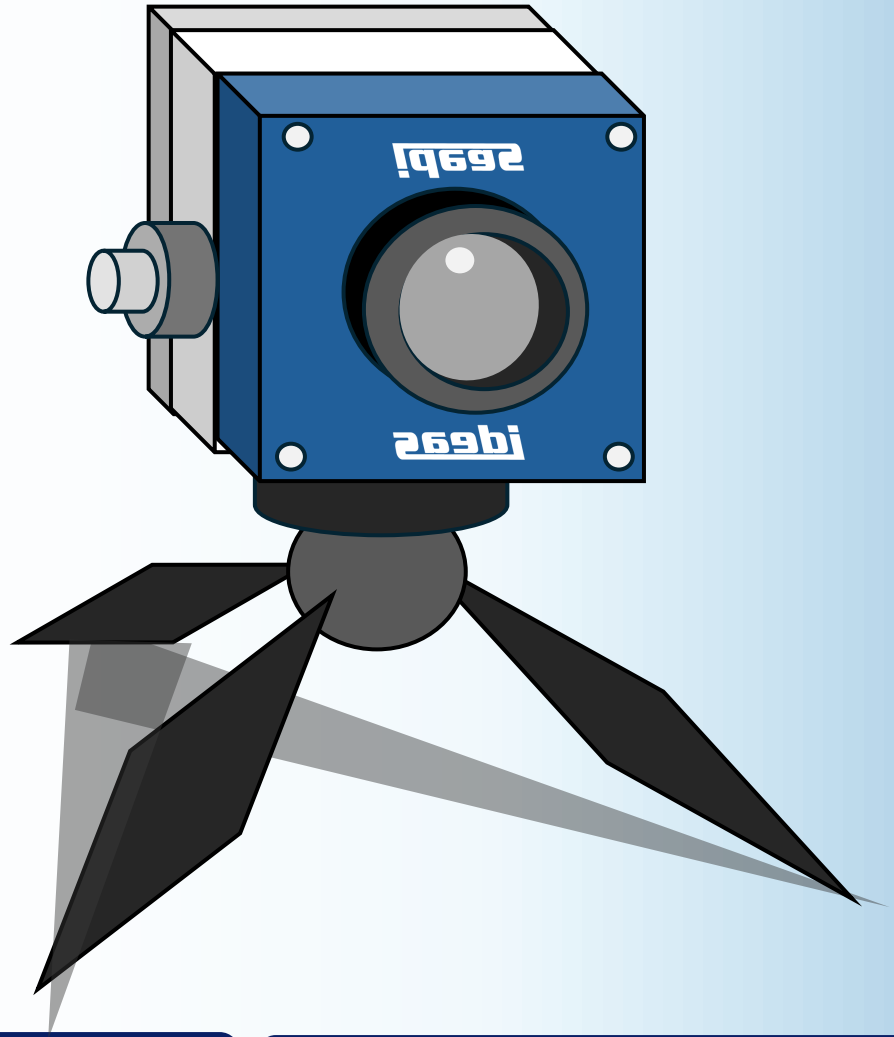
ideas

ossis

University of South-Eastern Norway





Embedded Calibration Storage and Drift Correction for an Uncooled Microbolometer Thermal Imaging System

Master Intern

Vu Hoang Nguyen

Background

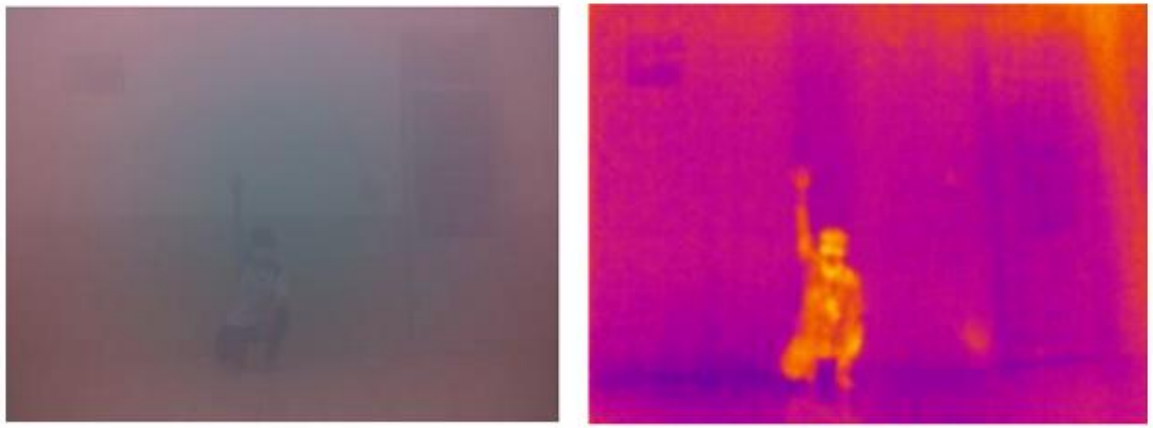


Figure 1: Normal camera compared to thermal camera in heavy smoke environment

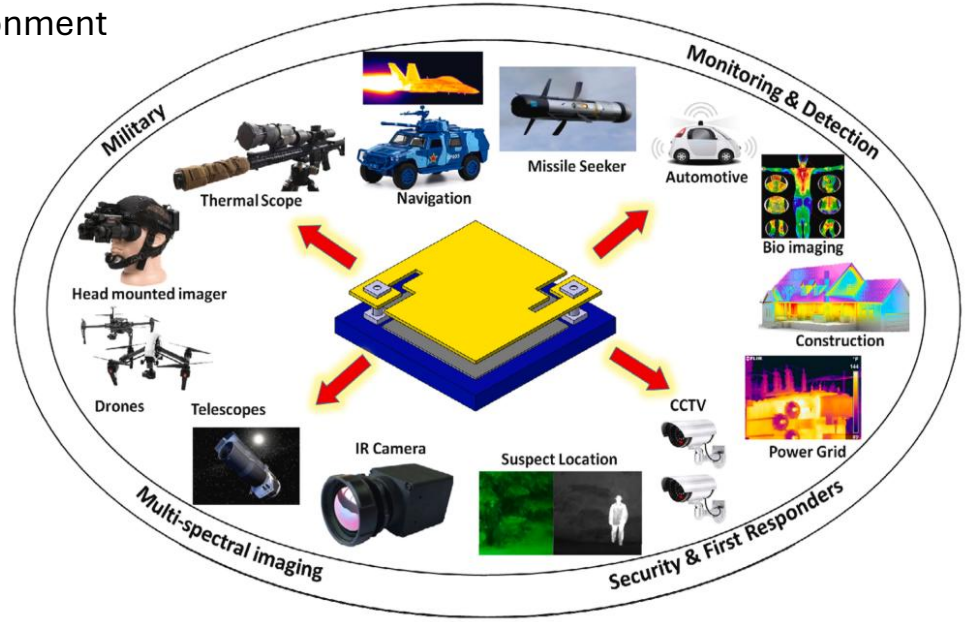


Figure 2: Application of uncooled IR microbolometers in various fields

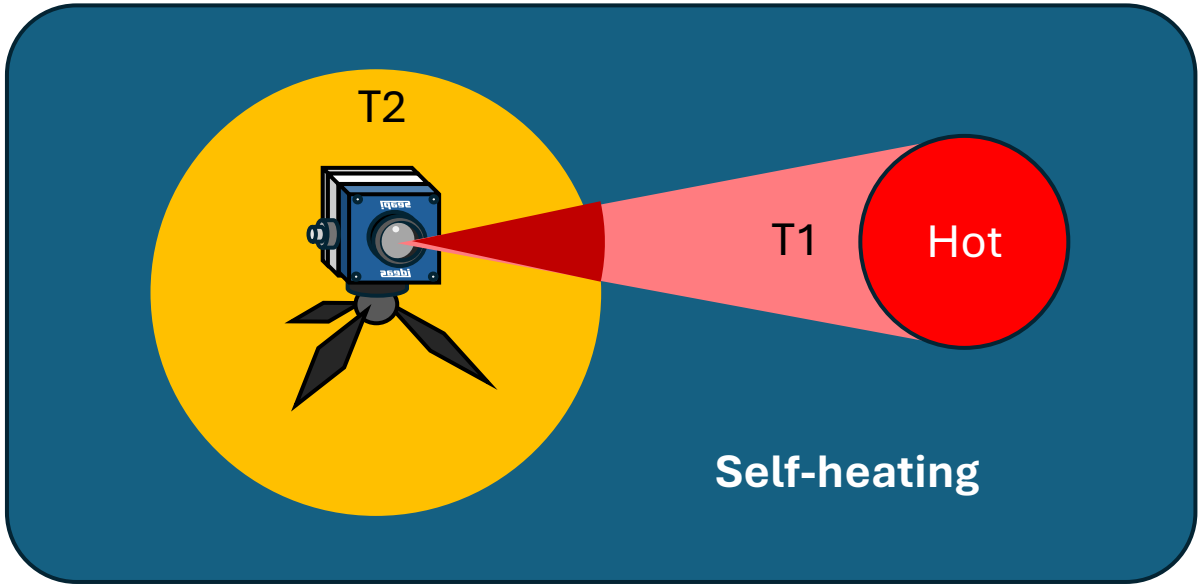


Figure 3: Self-heating issue of uncooled microbolometers detector

- ❑ Thermal camera are not affected by light change or low-light environment
- ⇒ It outperforms normal camera in many specific case like in heavy smoke area (Figure 1)
- ⇒ Thermal camra has been applied in many different field (Figure 2)
- ❑ However, it is suffered from self-heating issue: when the camera starts working, it also graduately heats up by itself, this heat also affect the data output (Figure 3)

Tasks overview

Goal

❖ Solve the issue with self-heating

Methodology

- Research on calibration method to enable working stability

1	One point shutter correction	QT desktop application: C++
2	Thermal drift correction	Jupytor notebook: Python
3	Embedded calibration storage	Vitis&Vivado: Embedded C

Overview of existing thermal camera system

Mobile System for Demonstrations

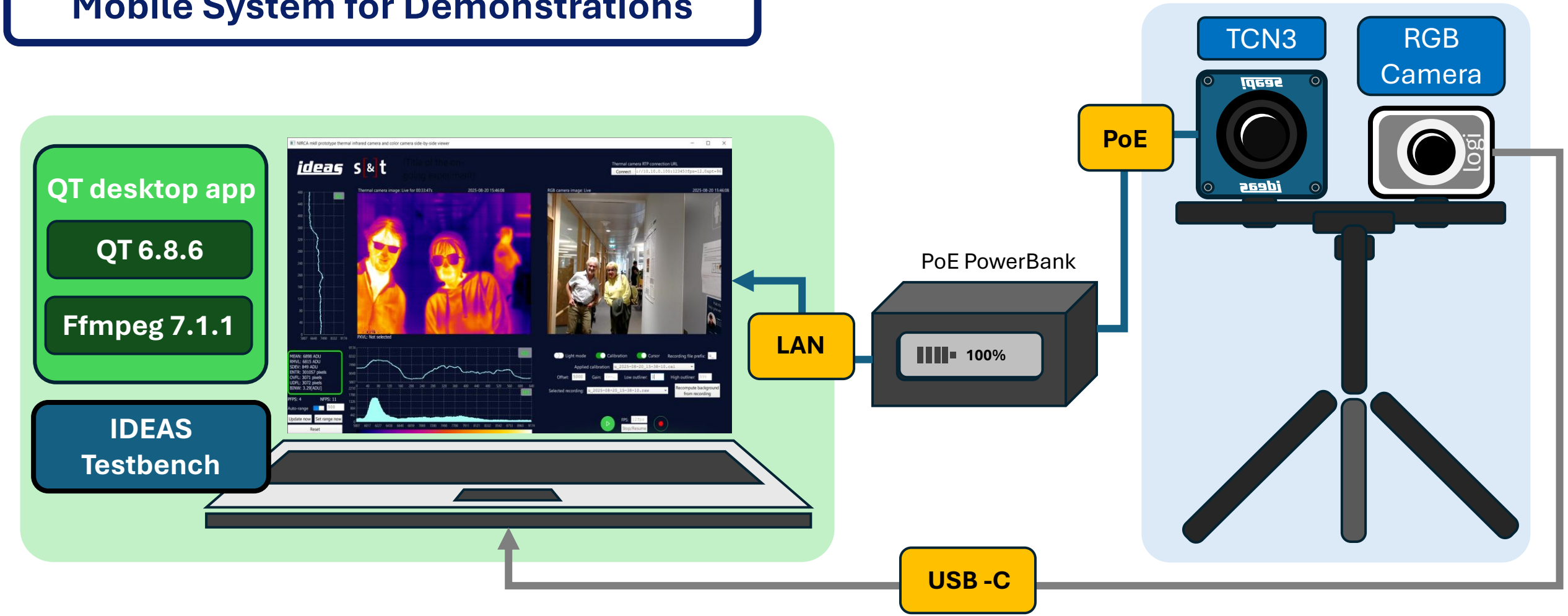
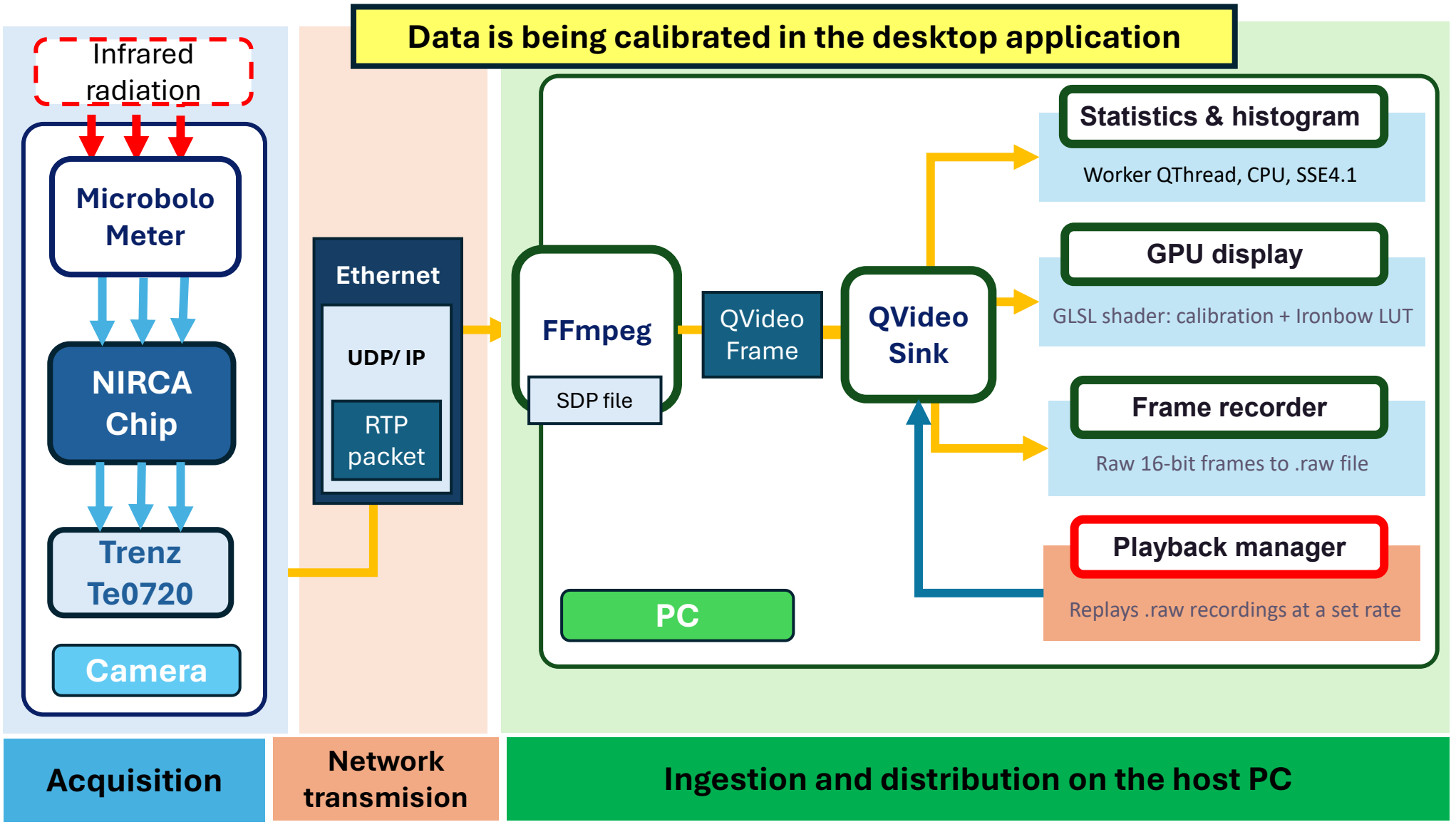


Figure 4: Thermal camera mobole system at IDEAS

Overview of the existing data flow



Acquisition
Infrared radiation warms each microbolometer pixel; the NIRCA chip digitises the array to 14 bits, and the TE0720 module packs the frames for the network.

Network transmission
One frame is 614 kB, far above the Ethernet MTU, so it is split across many RTP packets over UDP: low latency, with sequence numbers to reassemble it.

Ingestion and distribution
An SDP file generated at runtime tells FFmpeg to expect raw 16-bit greyscale. Every consumer then draws from one QVideoSink: display, statistics, recording, replay.

Overview of the Software user inteface

Software GUI

Graphs

Statistic

Histogram

Thermal and RGB camera

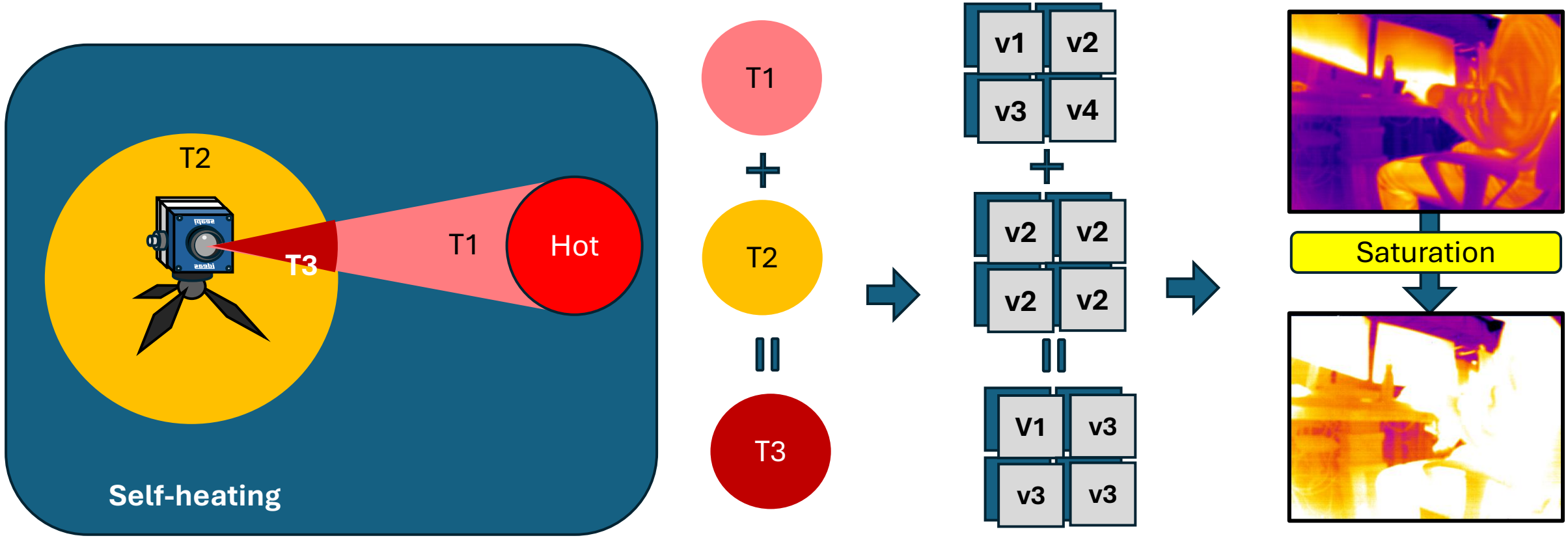
Set connection



Calibration and record

Issue with self-heating

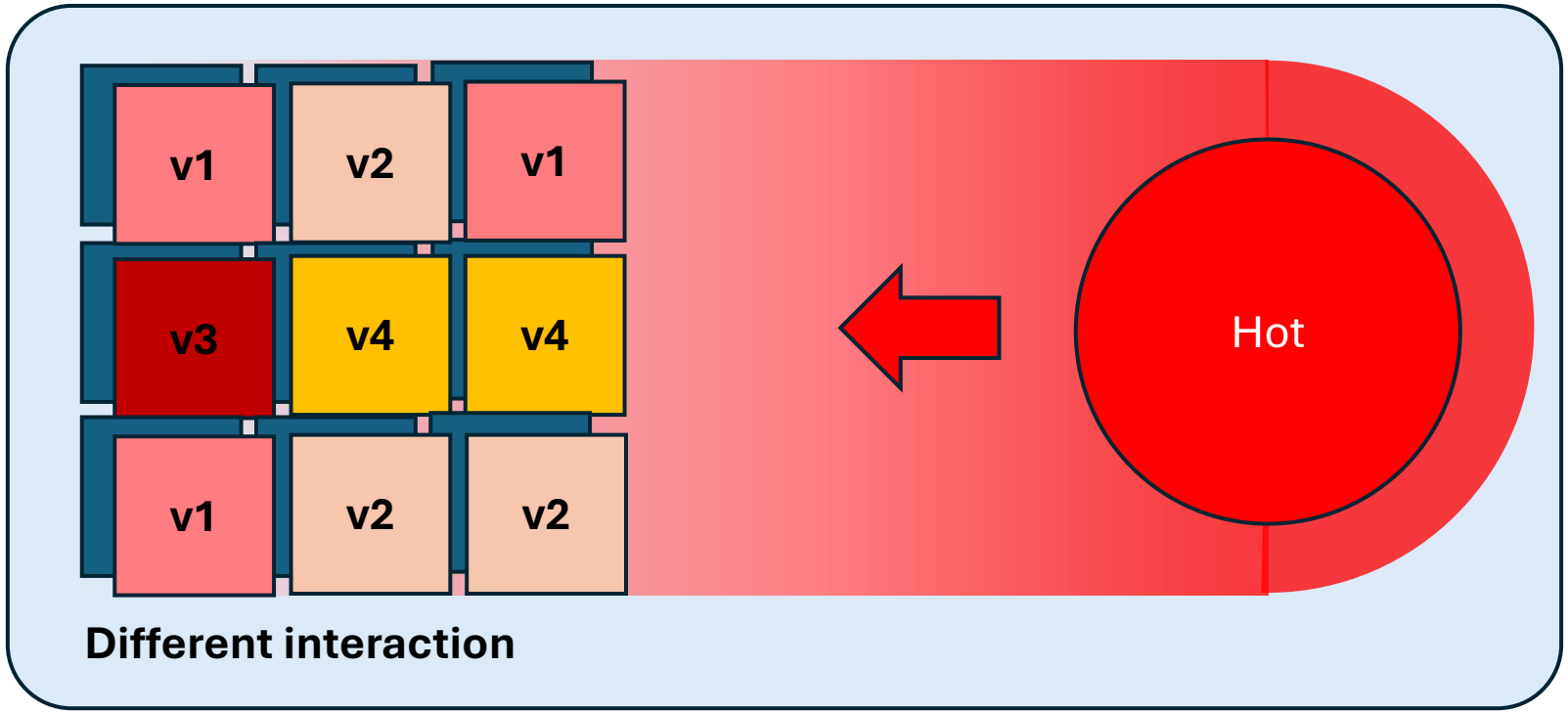
Heat from inside electronic circuit affect the total result

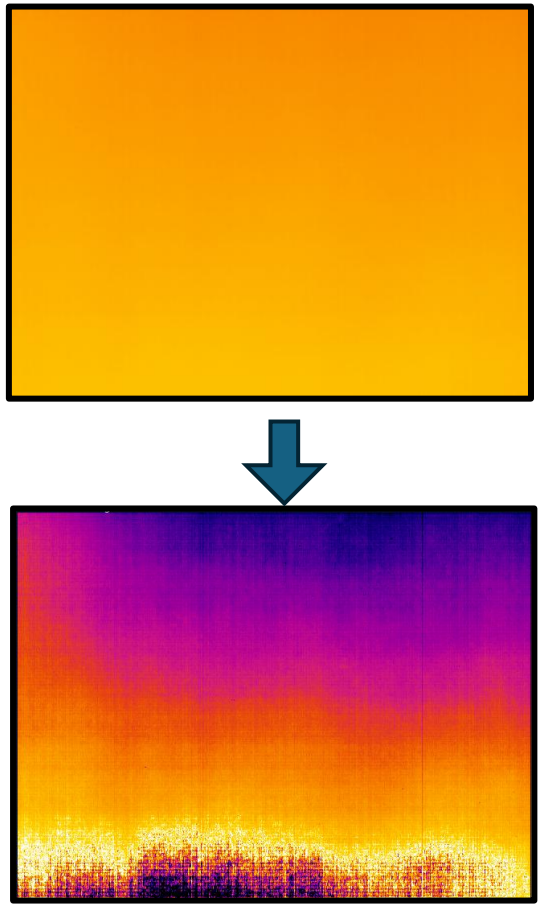


Instead of only reflect T1 of the scene, the output also afftect by T2 which is the heat released by the camera
T2 increases quickly after the camera starts working => pixel value becomes saturated (reach the maximum value while the other pixels is still increasing)=> After a while, a larger and larger parts become saturate (the white part) => lost data and details of the image.

Issue with self-heating

Each pixels react differently to the same amount of heat





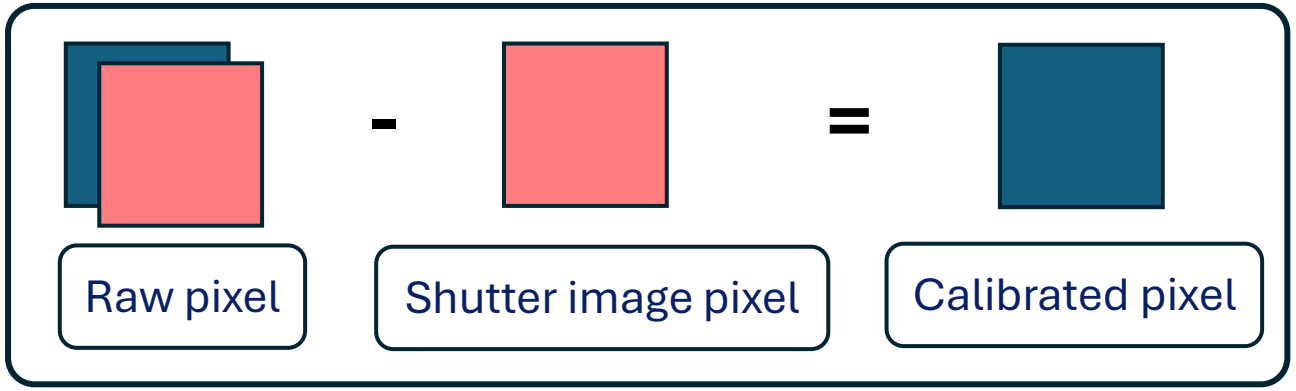
Each pixel reacts differently: with the same amount of heat, some increase value fast, some increase slower
=> Even at early state, the frame is uniform (all pixels have the same value), after a working period, because the camera continuely release heat and pixels react differently => the pixels value would be come different => frame graduately shows different color in different area

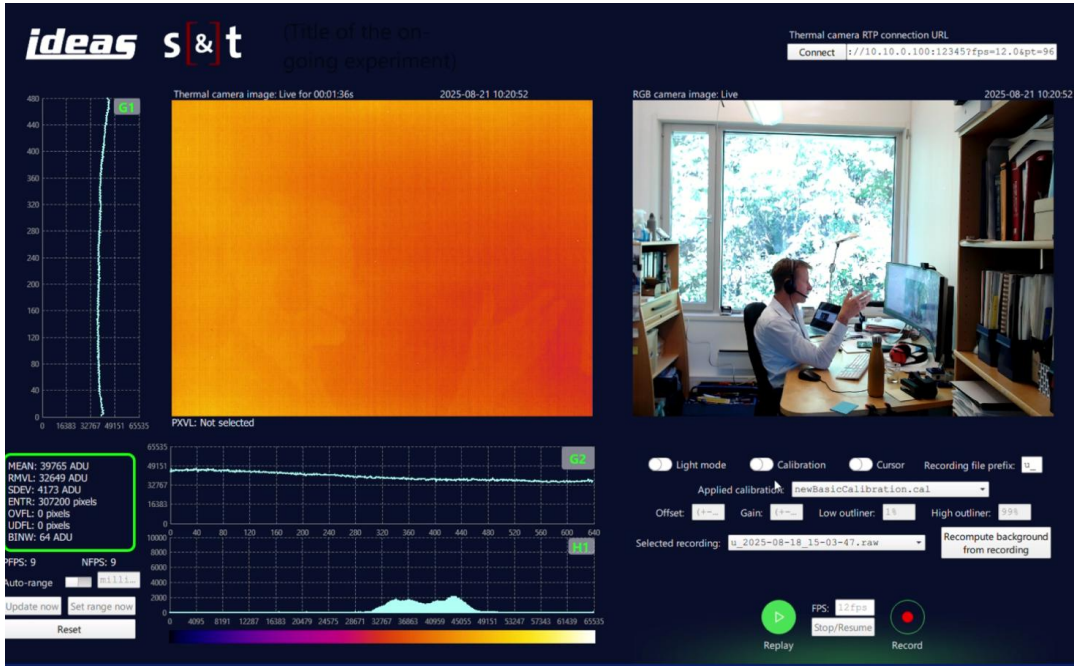
1 Point shutter correction

I 1 point Shutter correction

Basic principle

Using a shutter image as a reference



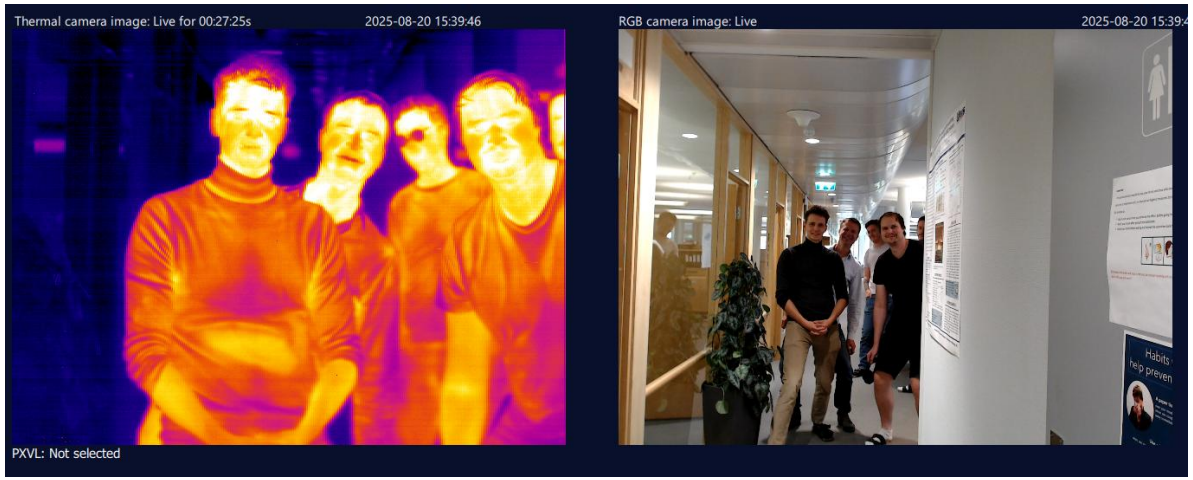


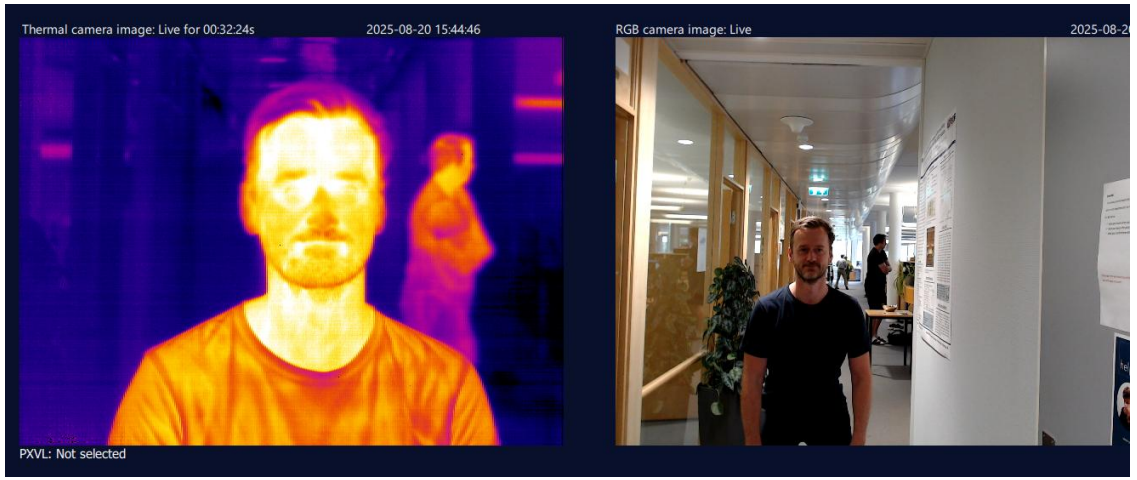


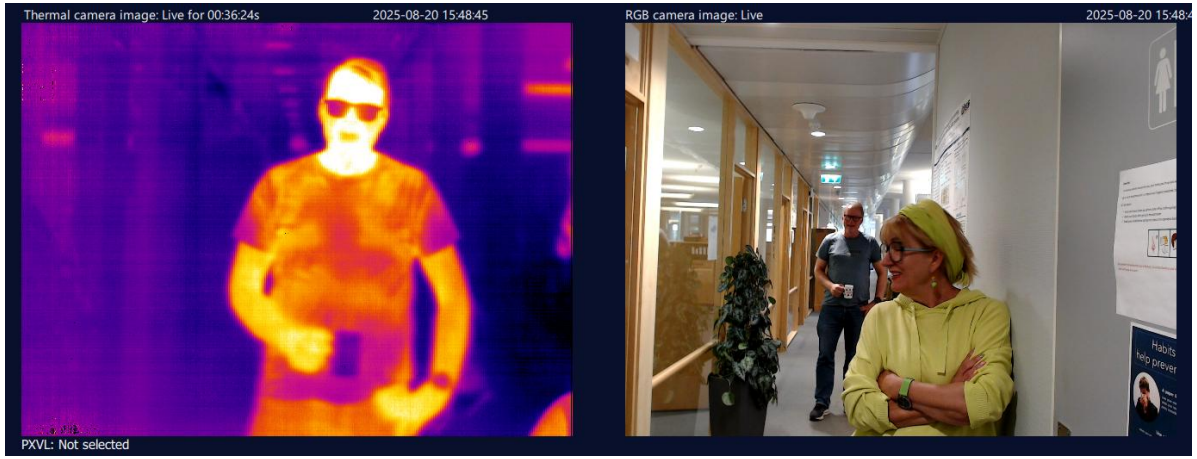
Before and after 1 point shutter correction

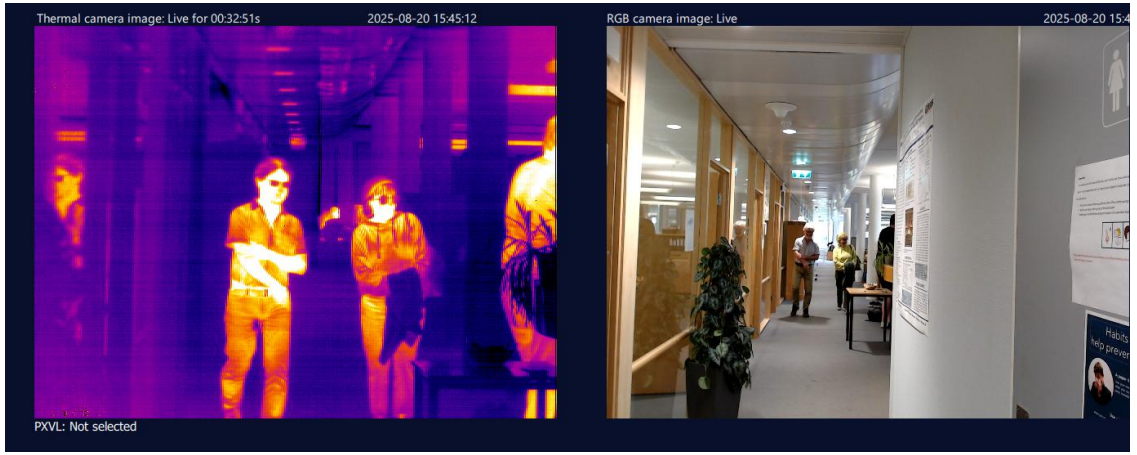
1 Point shutter correction

I Example results



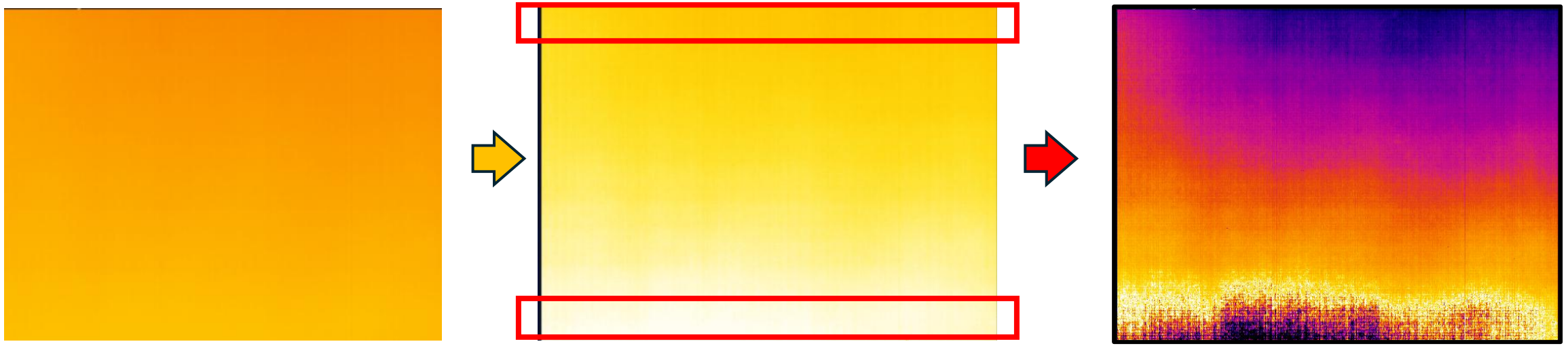






1 Point shutter correction

I Issue with 1 point Shutter correction



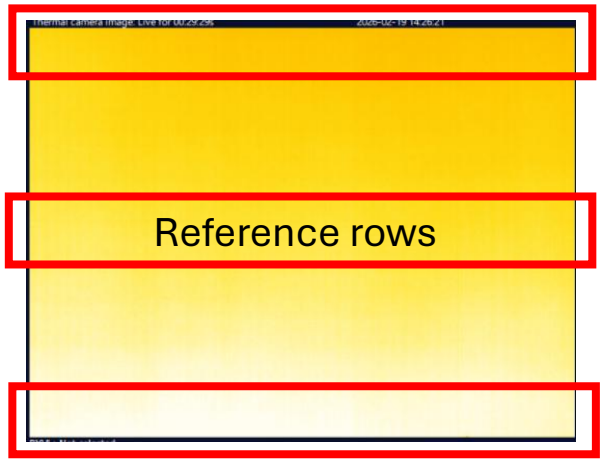
t = 0 (min) | t = 10 (min) | t = 50 (min)

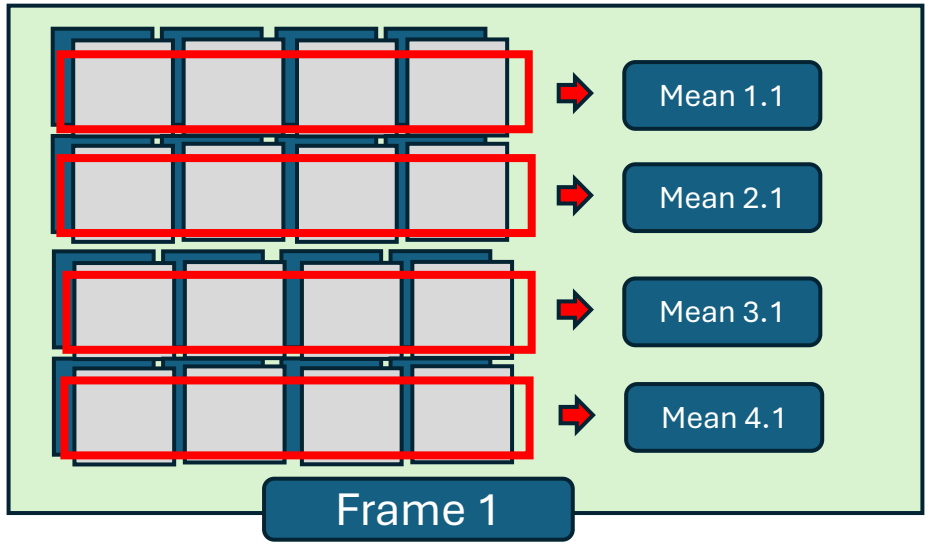
After applying the correction, the camera continues heating up which make the output data worse through time => need to do a new shutter correction after 3-5 minutes => interrupt a real-time video streaming continuously => not practical.

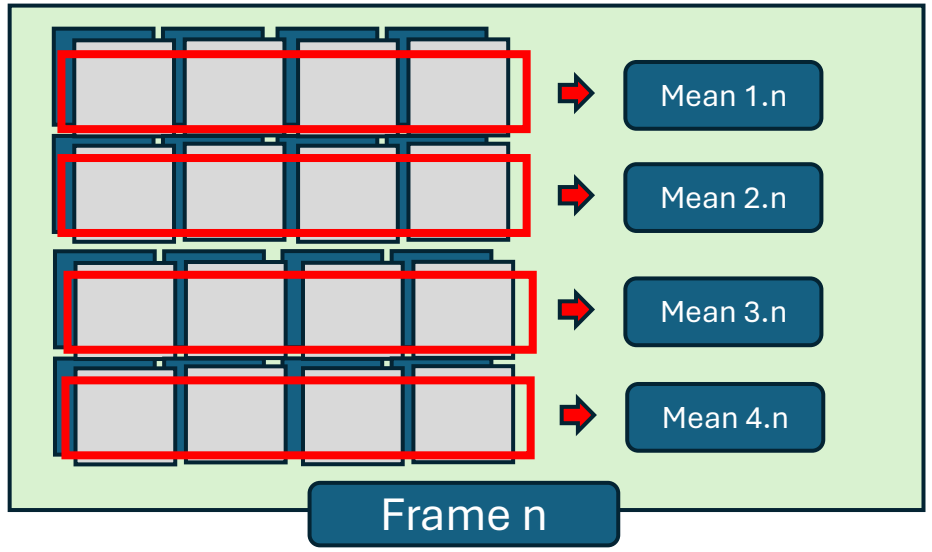
! **A noticeable point of thermal camera imaging system at IDEAS:** the upper rows of the frame always increase value slower and the bottom part always increase much faster. The middle ones seem to be increase graduately.

Row-level drift correction

II Row-level drift correction



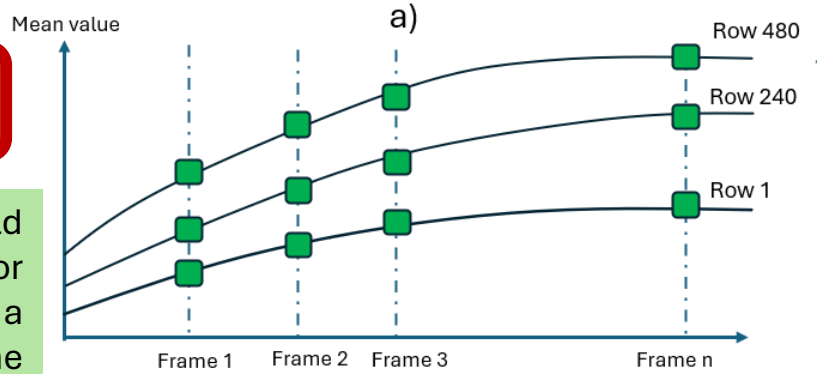




b)

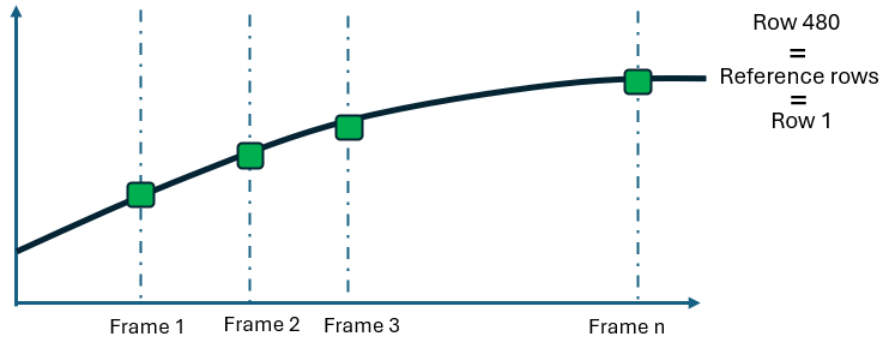
$$m(r,t) = A(r) \cdot \left(1 - exp\left(-\frac{t}{\tau(r)}\right)\right) + B(r)$$

Record a video from when camera cold to when it hot => build drift curves for each row => take the middle one as a reference => force them to have the same behavior with the middle ones.



a)

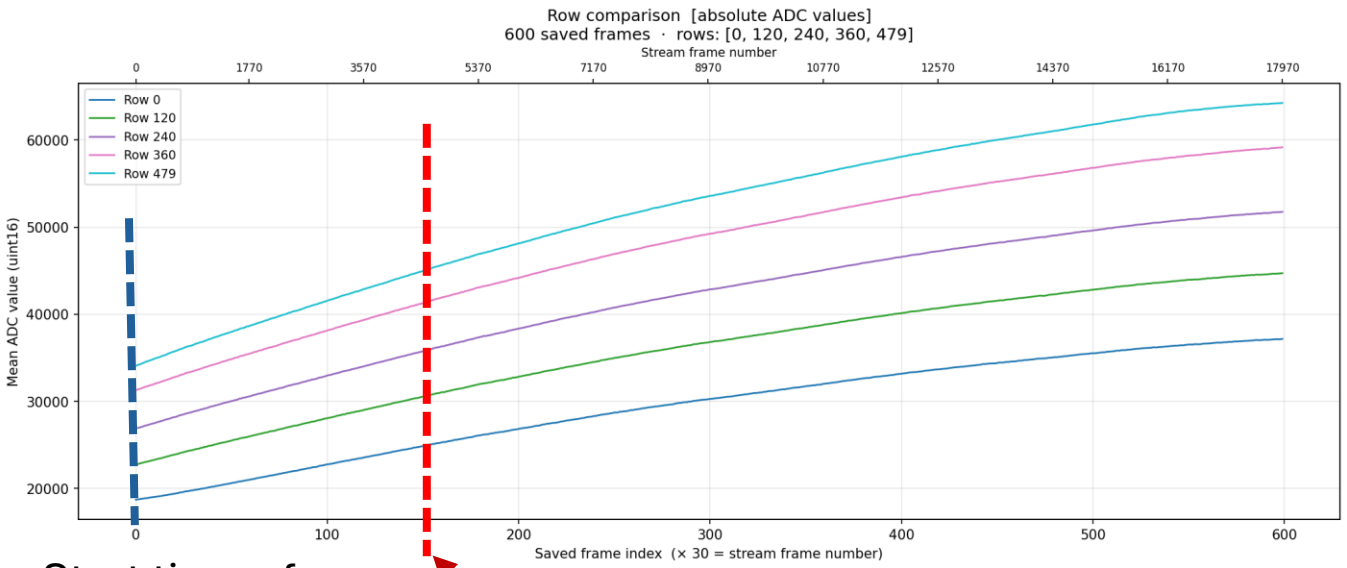
c)



d)

Row-level drift correction

II Issue when apply the correction with a new session



Start time of data source

Start time of Test video

When start a new session, time can be mismatch

The algorithm point out where is the right position

Using T (temperature) instead of t (time) depend

Not available as the current camera doesn't have a thermal sensor

Record a 10 second shutter video - > calculate 2 mean value at start and end -> define the position

Approach A

Apply a new shutter correction everytime start a new session and set the start time equal to start time of the data source

Approach B

Row-level drift correction

Experiment conducting

Experiment no.	Content	Camera	Camera temperature (°C)	Scene	FPS (fps)/ duration (minutes)
1	Source recording for drift fitting	TCN3-001	25	5	0.33 / 30
2	Test video 1	TCN3-001	25	10	11 / 10
3	Test video 2	TCN3-001	Room temperature (uncontrolled)	Unknown scene, 110-frame lens-cap calibration	11 / 30
4	Reproducibility source 1	TCN3-001	25	10	0.33 / 30
5	Reproducibility source 2	TCN3-001	25	20	0.33 / 30
6	Equipment-free source	TCN3-001	Room temperature (uncontrolled)	lens-cap	0.33 / 30
7	Source recording for drift fitting	TCN3-004	20	5	0.33/20
8	Test video 2	TCN3-004	Room temperature (uncontrolled)	Unknown scene, 110-frame lens-cap calibration	11/10

Controlled-condition experiment: Experiment 1 as the data source and experiment 2 and 3 as the test video.
Purpose: verify the performance of the approach with known condition

Uncontrolled-condition experiment: Experiment 6 as the data source and experiment 3 as the test video
Purpose: verify if the approach can work in any condition as the laboratory tools are expensive and required more afford

Reproducibility experiment: Experiment 1,4 and 5 have the same condition recorded in different session and experiment 2 as the test video.
Purpose: verify if the approach shows similar result with different data source

Experiment 7 and 8 replicate controlled-condition experiment but with different thermal camera prototype.
Purpose: Verify if the approaches shows similar result with different camera prototype

Row-level drift correction

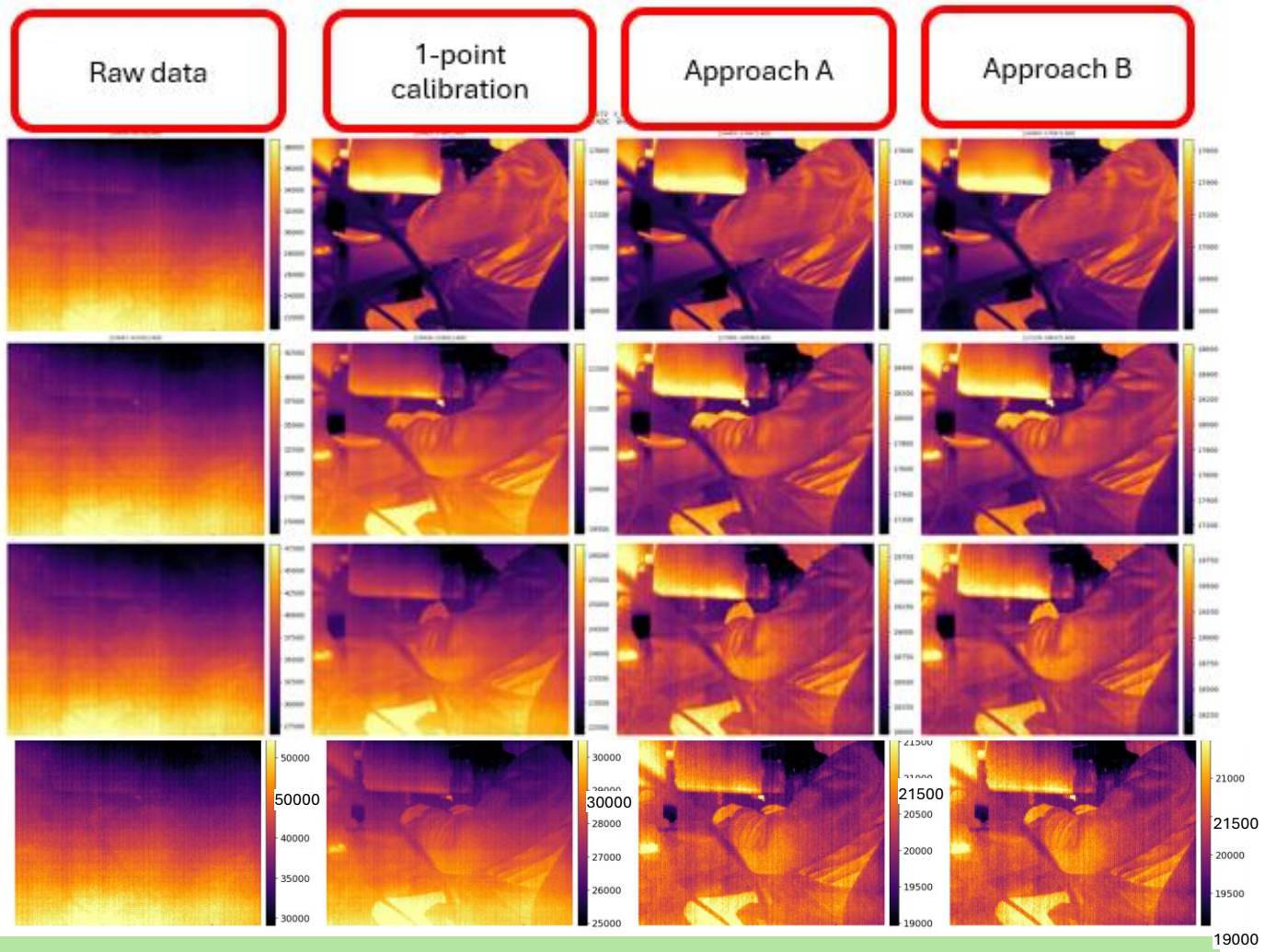
Condition-controlled experiment

Data source:

- Record 600 frame within 30 minutes (3 second / frame)
- Camera start temperature: 25 (°C)
- Scene: blackbody at 5 (°C)
- Require laboratory tool: thermal chamber and black body.

Test video:

- 10 minutes video
- Camera start temperature: room temperature



Approach A and B are almost similar, outperform 1-point shutter calibration though time. At t = 520(s) images given by approach A and B show less gradient, keep details at the upper and lower parts of the frame

Row-level drift correction

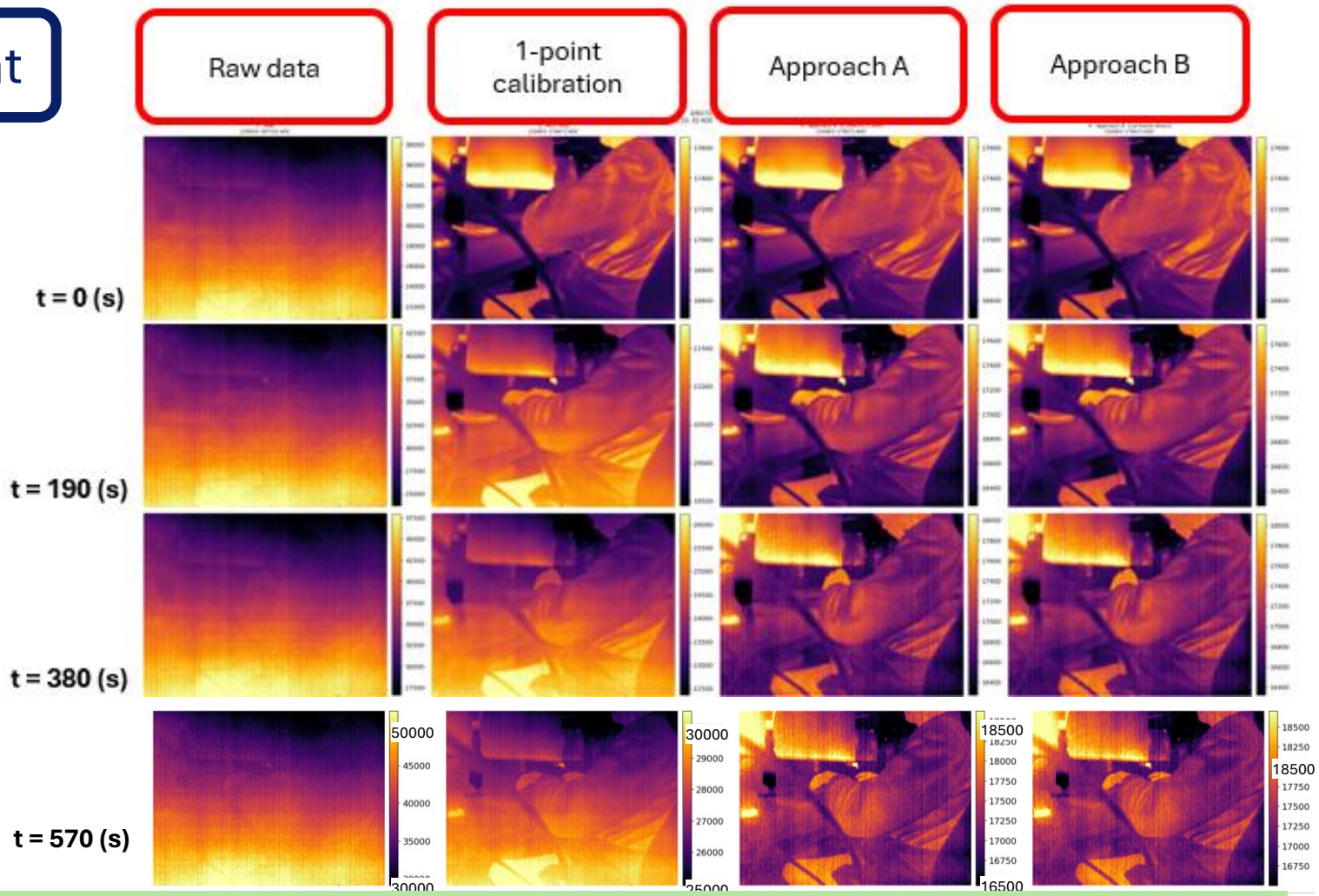
Condition-uncontrolled experiment

Data source:

- Record 600 frame within 30 minutes (3 second / frame)
- Camera start temperature: room temperature
- Scene: A lens cap
- Purpose: verify if this approach can stil usable without laboratory tools and evironment

Test video:

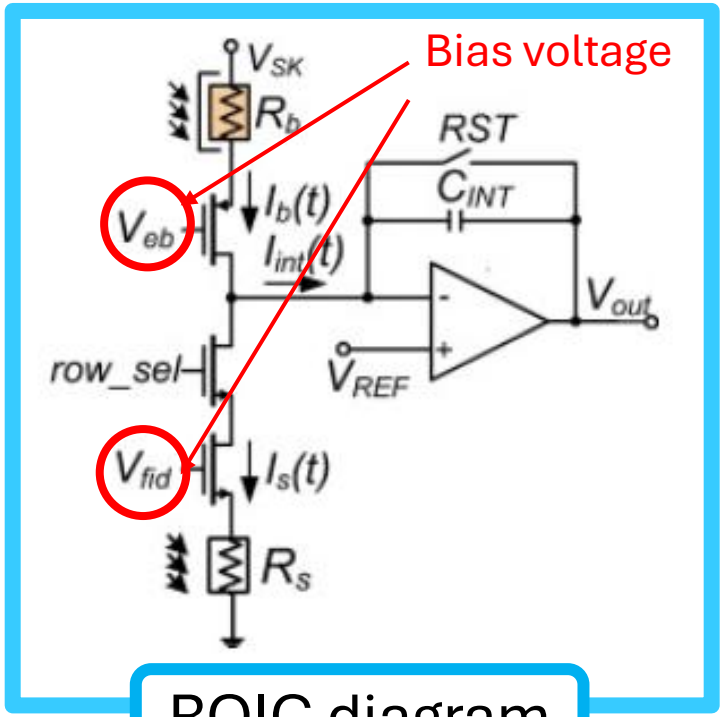
- 10 minutes video
- Camera start temperature: room temperature



Similar to last experiment, the image at t =570s of Approach A and B are outperform the one by 1-point calibration. The gradient of the frames are decreased more than the condition-controlled experiment

Embedded Calibration Storage

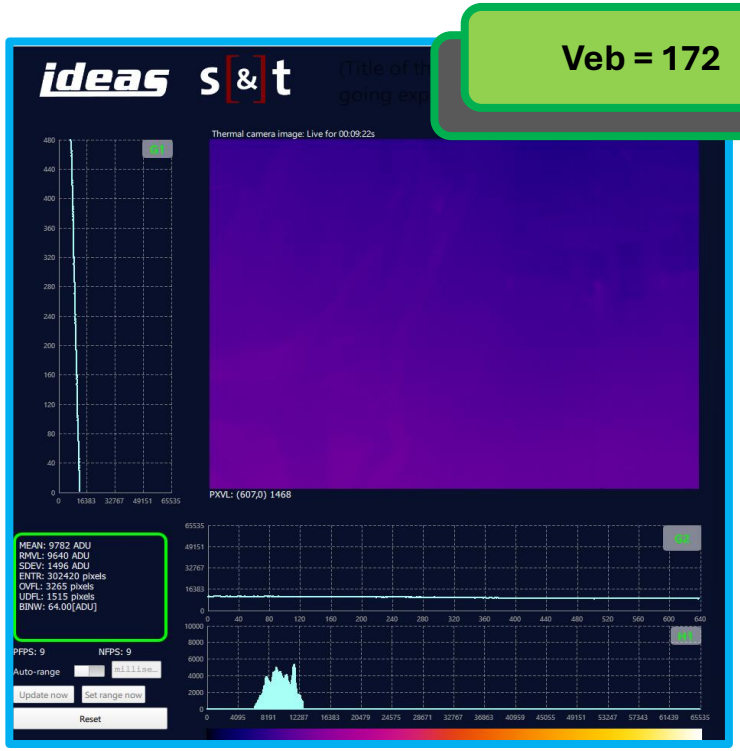
III Calibration storage



ROIC diagram



(Value range: 20000 - 32000)



(Value range: 6000 - 12000)

By changine Veb and Vfid (Bias voltage of active microbolometer and blind microbolometer) all the pixel value can be shifted backward or forward.
Eg: at Veb, the pixel value range is 20000-32000 but change to 6000-12000 when decrease Veb by 2 value.

Idea: when the pixels are almost saturated => change (Veb, Vfid) to move the range backward => slow down the saturation

Embedded Calibration Storage

III Calibration storage

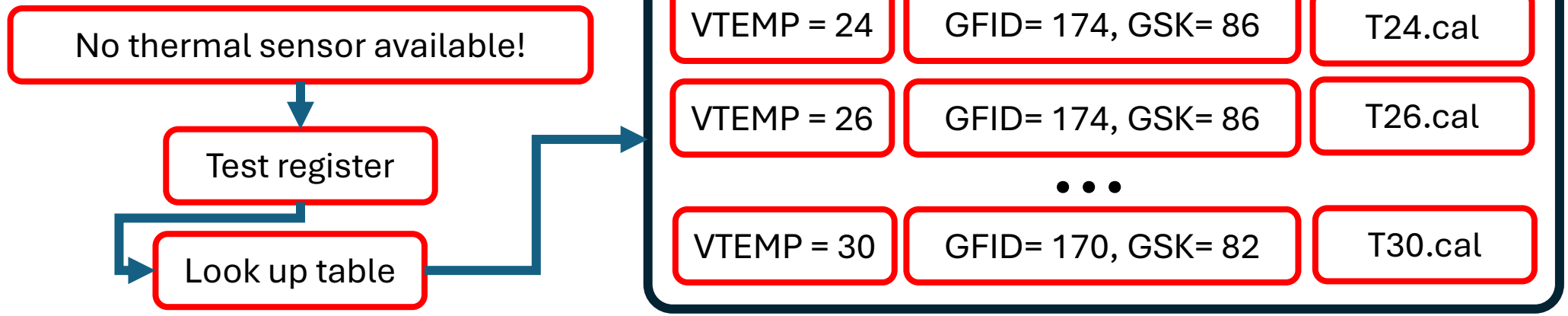
Idea

- Record shutter calibration at certain temperature
- Apply stored shutter calibration after 2 degree
- Change set of {Veb;Vfid} every 10 degree

→ Ensure sharp and clear image

→ Slowdown saturation

→ Embedded code in Vivado & Vitis



Calibration inside the camera

INSIDE THE CAMERA · ZYNQ-7000

eMMC
calibration files in /lut/cal/

→

Lookup table
selects by substrate temperature

→

Per-pixel offset
subtracted line by line

→

RTP out
frames leave already corrected

Ethernet

HOST PC

Receives corrected frames
No calibration work left for the host

The correction is applied to the detector output before transmission, not after it.

1 Enable the eMMC

The eMMC sits on the SD1 controller through MIO bank 501, which the original design never routed. The Vivado project was reconstructed from its TCL script, SD1 and its MIO pins enabled in the PS configuration, and a new hardware description exported. In firmware, the BSP gained xilffs, coupling FatFS to the Zynq SD/eMMC driver, with a mount routine at start-up.

2 Move files onto the device

An FTP service inside the network module: authentication, directory listing, upload and download, with passive-mode data transfer for compatibility with common clients. Verified against WinSCP and FileZilla.

3 Select by temperature

A lookup-table module reads the bias and calibration tables from the eMMC. The calibration file changes every 2 °C of substrate temperature and the GFID and GSK bias values every 10 °C, with hysteresis so the selection does not oscillate at a threshold.

4 Apply on board

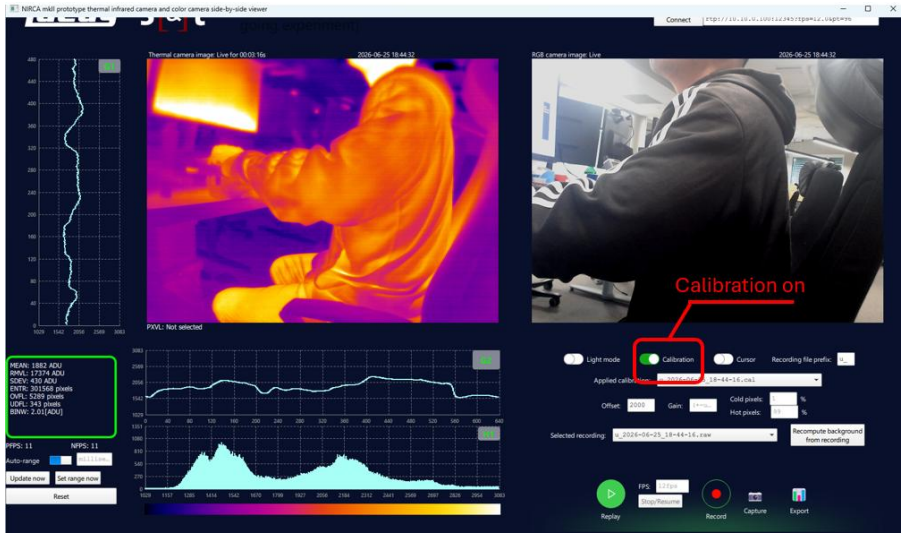
One offset per pixel, subtracted line by line as the frame is read out, so no second copy of the frame is needed. Double-buffered, so a new file loads while the pipeline keeps reading, and an identity calibration takes over if a file is missing, so the video never stops.

VERIFIED A test register forced two substrate temperatures. Bias values were held equal across both and host-side calibration was switched off, leaving the calibration file as the only difference. Switching codes produced the corrected image, so selection and correction both ran inside the camera.

Bias and calibration tables currently hold preliminary values; the 2 °C and 10 °C intervals are not yet measured.

Embedded Calibration Storage

III Calibration storage



Applied t20.cal file by software

Move into eMMC

Change test register to 200

(Vfid,Veb): (172, 78) to (172, 86),

Set (Vfid,Veb) back to (172,78)



Applied t20.cal file by software

Testing to verify the technical part of the approach

Summerize

Tasks has been done

- Implemented 1 point shutter correction
- Developed a row-level drift correction method and conducted different experiment to verify the efficiency
- Move the calibration phase from QT desktop application side to the firmware side
- Developed a calibration storage approach and verify the working

Thank you!

